Readme for External Structures

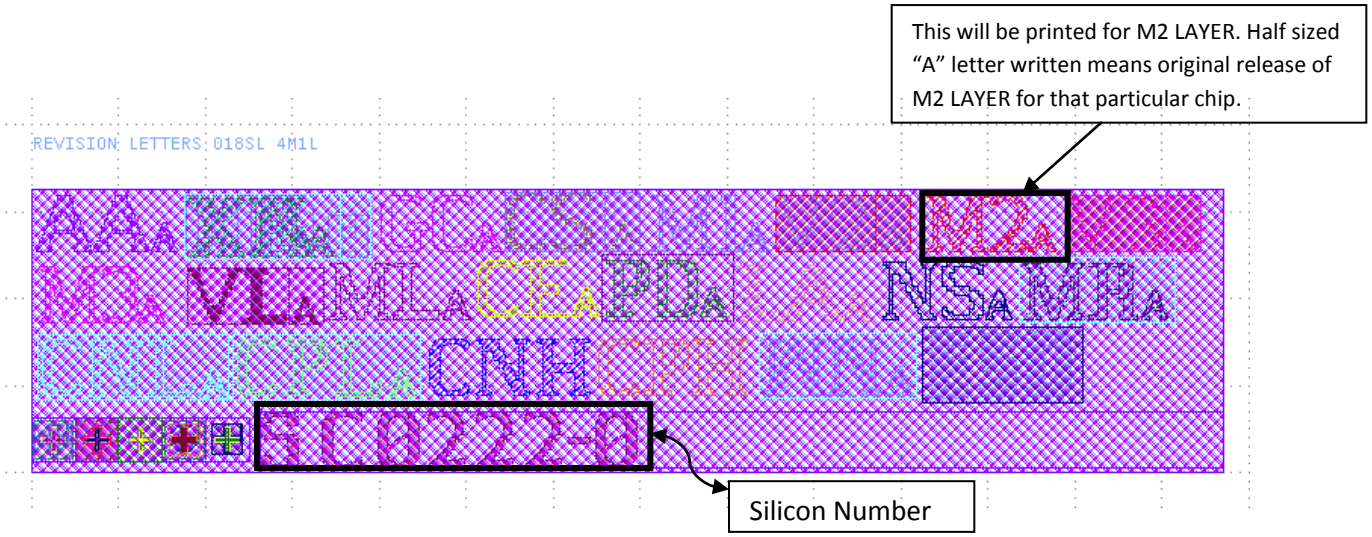
(SCL180SL process)

1. Introduction

External structures like Silicon number, revision block are required to be put in every individual chip design for chip identification, layer revision control. In addition to these there are other structures like seal ring and PADs which are to be used by designer to put at appropriate positions. There are four different process options viz. 4M1L, 4M1T, 6M1L and 6M1T based on the number of metal and last metal. Brief Description of the structures is as follows:

Revision Block: Every device must have Layer names and revision letters digitized on mask. If the Layout digitized on a mask has been modified, the revision letter for that Layer must be changed. Without Layer name and revision it will be difficult to identify after assembly and during Mask View, which mask revisions were used for a specific product.

Revision Letters are available in several options depending on flavor of design and technology.



Silicon Number: It is the distinct number/name for every chip or design for its identification at every stage. Silicon number is provided by SCL DCC.

Seal Ring: Seal ring structure is disclosed for protecting the core circuit region of an integrated circuit chip. The seal ring structure includes a metallization layer, having a bridge sub level and plug sub level (*Refer 2.2. Seal Ring on page no. 4).*

2. Guidelines for preparing and inserting External Structures in the design.

Path of All GDSII:- /usr1/dtech/scltech/scl180/ext_str directory

Information of GDSII: Following are the different GDSII for different purposes explained below:

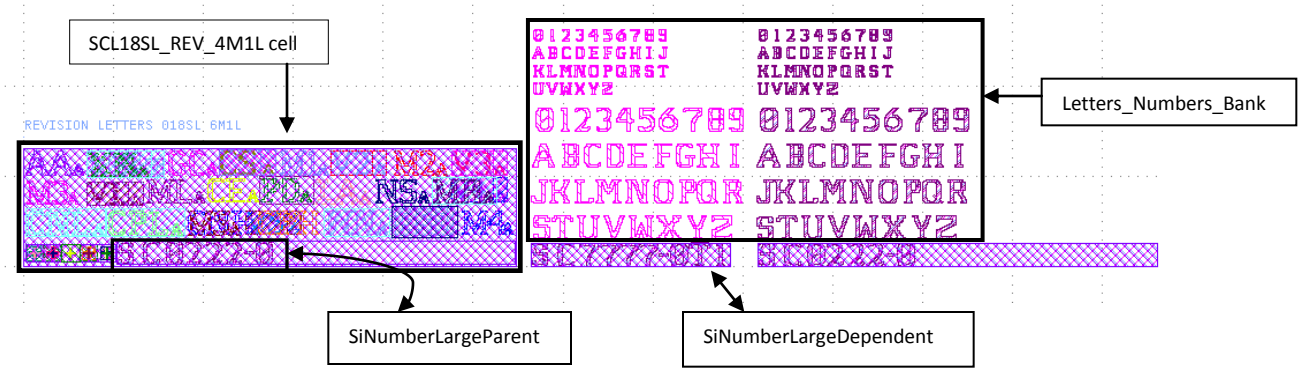
1. SCL18SL_4M1L_PDK2020.gds Top Cell: SCL18SL_4M1L
   GDS MD5 checksum: 5e0dfc90729d2a030aee96c8033467a3
2. SCL18SL_4M1T_PDK2020.gds Top Cell: SCL18SL_4M1T
   GDS MD5 checksum: f68786342f48c6229ae5286730bb46b5
3. SCL18SL_6M1L_PDK2020.gds Top Cell: SCL18SL_6M1L
   GDS MD5 checksum: f717925b238c48720af814690206207b
4. SCL18SL_6M1T_PDK2020.gds Top Cell: SCL18SL_6M1T
   GDS MD5 checksum: 90f11c51c71cc9cdcbd34de178c57d58
5. SEALRING_PAD_PDK2020.gds Top Cell: Top_SCL18SL_SEALRing_Pads
   GDS MD5 checksum: 787ca57df16e1d3bfc2f5030e560c269
6. Rev_Let_holes.gds Top Cell: Rev_Let_holes
   GDS MD5 checksum: a3c24752afebd4b80d98fa085f6e3248

2.1 Silicon Number & Layer Name and Revision Block

a. Get the Silicon number issued for all type of designs from SCL DCC. For External designs other than SCL, user need to contact Mask division (skverma@scl.gov.in & vandana@scl.gov.in) to get the silicon number issued after getting project approval from SCL's Project Planning Group.
b. Designer has to choose appropriate GDSII from the above mentioned list as per their design. There are four options available as per metal scheme in GDSII.

A. 4M1L

For this option user need to choose **SCL18SL_REV_4M1L** cell from SCL18SL_4M1L_PDK2020.gds.



B. 4M1T

For this option user need to choose **SCL18SL_REV_4M1T** cell from SCL18SL_4M1T_PDK2020.gds.

C. 6M1L

For this option user need to choose **SCL18SL_REV_6M1L** cell from SCL18SL_6M1L_PDK2020.gds.

D. 6M1T

For this option user need to choose **SCL18SL_REV_6M1T** cell from SCL18SL_6M1T_PDK2020.gds.

COMMON cells under above mentioned four cells are as follows with their purpose:

Letters_Numbers_Bank: Bank of letters and numbers to be used to prepare the revision block and the silicon number as described below.

SiNumberLargeDependent: Used when the Silicon number is for a test circuit design.

SiNumberLargeParent: Used for the parent design.

c. Prepare the Silicon number as required by replacing and adding the cells for individual letters and numbers by the cells for individual letters and numbers within the cell 'Letters_Numbers_Bank'. The cells for these letters have names 'char_lA' for the letter A, 'char_l0'* for the number 0 and so on.

d. Choose the ANY ONE gdsii 'SCL18SL_4M1L_PDK2020.gds / SCL18SL_4M1T_PDK2020.gds / SCL18SL_6M1L_PDK2020.gds / SCL18SL_6M1T_PDK2020.gds ' to the design depending upon the metal option in the design. It should be placed preferably at the bottom left of the design.

e. Do not remove any support Layer (i.e., DEV_AREA_ext (77/65), AREA10_opc (80/30), ACTIVE_nodummy(9/35), NODUMMY(81/0) etc.) from any of the cell of any GDSII.

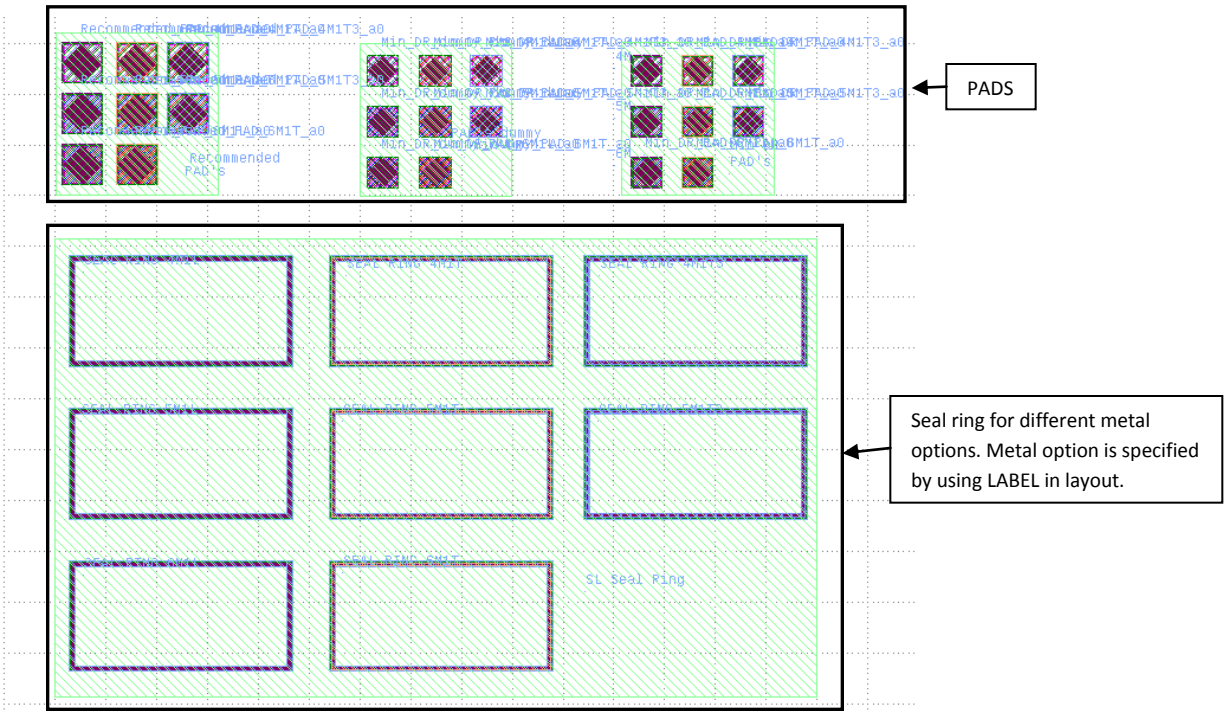
f. No electrical structures should cross these support layers. Mainly Revision block and silicon number must be placed at a safe distance from circuit area of design. Please refer to the relevant Design Rule document for other related rules.

g. In the cell 'SCL18SL_REV_4M1L / SCL18SL_REV_4M1T / SCL18SL_REV_6M1L/ SCL18SL_REV_6M1T ' replace relevant layer revision letters by the appropriate half-sized letters(see figure at first page) from the cell 'Letters_Numbers_Bank'. Use the letters drawn using the GC/lc layer and change them to the appropriate layer in data type zero.

2.2 Seal Ring

Seal Ring is available in eight options: 4M1L, 4M1T, 4M1T3, 5M1L, 5M1T, 5M1T3, 6M1L and 6M1T.

Required Seal ring can be picked up based upon the metal options from the above mentioned SEALRING_PAD_PDK2020.gds GDSII.



2.3 Pads

The designer is advised to use SCL PAD design for smooth DRC. User need to refer SEALRING_PAD_PDK2020.gds

There are 3 optional Types of Pads: Minimum design rules Pads, Recommended Pads and Minimum design rules dummy Pads.

Each of the 3 Types is available in 8options of metal combinations and suitable for SL technology.

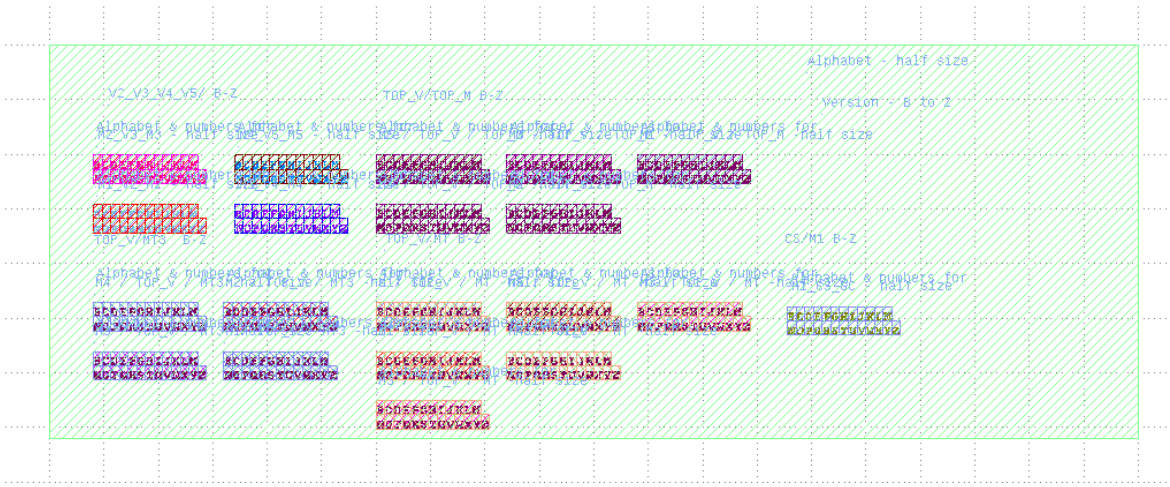
1. Metals size of minimum design rules Pads is 60ux60u. Pad also contains Layer DEV_area(pad) 77dt46 at size 60ux60u and Layer PAD(lc) 68dt0 at size 57ux57u. *(Sub-cell: PAD_min_DR)*

2. Metals size of recommended Pads is 80ux80u. Pad also contains Layer DEV_area(pad) 77dt46 at size 80ux80u and Layer PAD(lc) 68dt0 at size 75ux75u. *(Sub-cell: PAD_RECOM)*

3. Metals size of minimum design rules dummy Pads is 60ux60u. Dummy Pad also contains Layer PAD(dy) 68dt8 at size 60ux60u. *(Sub-cell: PAD_dummy_min_DR)*

2.4 Rev_Let_holes

This sub-cell contains the small (half sized) numbers and alphabets for revision change in the contact and via layers for different metal schemes (Refer GDSII).



NOTE:

In case of any query regarding anything related to GDSII and placement of external structures, please contact skverma@scl.gov.in & vandana@scl.gov.in .